R Chahana

Integrated M.Sc. Data Science Student | Amrita Vishwa Vidyapeetham, Coimbatore, India

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LinkedIn | Google Scholar | HackerRank

Research Profile

Final-year Integrated M.Sc. Data Science student with a strong foundation in probability, statistics, linear algebra, optimization, and machine learning. Interested in the mathematical foundations of deep learning, statistical learning, and probabilistic methods.

Education

2022 – 2027 | **Amrita Vishwa Vidyapeetham, Coimbatore, India**
Integrated M.Sc. Data Science (final year) — CGPA: 8.91/10

2021 – 2022 | **Gedee Public School, Coimbatore, India**
Class XII (CBSE), Mathematics–Physics–Chemistry with Computer Science — 87.17%

2020 | **Gedee Public School, Coimbatore, India**
Class X (CBSE) — 92.2%

Publications

- Kirubavathi Ganapathiyappan, **Chahana Ravikumar**, Raghul Alagunachimuthu Ranganayaki, Ayman Altameem, Ateeq Ur Rehman, Ahmad Almogren. Advancing Android Ransomware Detection with Hybrid AutoML and Ensemble Learning Approaches. *Computers, Materials & Continua (Tech Science Press)*, 87(1), Article 27, 2026. (link)
- Venkateswaran Radhakrishnan, **R Chahana**, Aarthi S, Kirubavathi G. An AI-Based Attention-Enhanced BERT and BiGRU Hybrid Model for Twitter Sentiment Analysis. *Proceedings of the 2026 International Conference on Computational Intelligence Systems and Devices (ICCISD)*, IEEE, Sharda University, 2026. (link)

Research Projects (ongoing)

Sep 2026 - Present | **Master's Thesis: Benchmarking Unsupervised and Deep Learning Approaches for UAV Flood Segmentation, with Visual Question Answering for Disaster Assessment**

- Extending an unsupervised flood-segmentation baseline from binary to multi-class segmentation (flooded buildings and roads) for annotation-free, low-computation processing.
- Benchmarking against conventional deep learning models using F1, IoU/Dice, boundary metrics, parameter count and inference time, with cross-dataset validation
- Determining when an unsupervised pipeline suffices and when deep learning justifies its compute cost; adding a Visual Question Answering layer for natural-language queries.

Jan 2026 – Present | **A Stacked Ensemble Framework for Multilingual Hope Speech Detection in Low-Resource Dravidian Languages** (*Manuscript in preparation*)

- Developed a stacked ensemble combining XLM-RoBERTa, mBERT, ELECTRA and BiGRU (GloVe embeddings), with XGBoost as the meta-learner.
- Addressed severe class imbalance with back-translation data augmentation and ensemble-guided filtering of synthetic samples.
- Extended the framework to low-resource Tamil via cross-lingual transfer learning and used LIME for interpretability.

Selected Projects

May 2026 | **Physics-Informed Neural Networks for State-to-State Learning of Nonlinear Stress Evolution in Cyclic Plasticity**

- Developed a state-to-state PINN that learns the one-step evolution of replacement stress in cyclic plasticity without time integration at inference.
- Formulated training losses using ODE residuals, multi-step rollouts, and initial conditions, with Adam and L-BFGS optimization.
- Validated against a Runge–Kutta reference, reaching 0.4 MPa RMSE (0.085 MPa MAE).
- *Tools:* Python, PyTorch, Adam, L-BFGS, NumPy, Matplotlib.

Mar 2026 | **Advancing Multi-Day Weather Forecasting using Temporal Deep Learning Networks**

- Developed a hybrid GA-ConvLSTM framework to forecast temperature, humidity, dew point and wind speed from 15 years of meteorological data.
- Combined CNN-based spatial feature extraction with LSTM temporal modeling and used transfer learning for multi-city adaptation, achieving R^2 of 0.971 for Chennai and above 0.98 for other cities.
- Optimized key hyperparameters with Genetic Algorithms to improve generalization across climatic datasets.
- *Tools:* Python, TensorFlow, Genetic Algorithms, Transfer Learning, Scikit-learn, Matplotlib.

Mar 2025 | **Fatty Liver Disease Detection from Ultrasound Images using ML/DL**

- Developed deep learning and classical machine learning models to detect fatty liver disease from ultrasound images.
- Performed image preprocessing and exploratory analysis using normalization, PCA and t-SNE.
- Implemented CNN models and baseline classifiers (Random Forest, Logistic Regression, Decision Tree), achieving 82% classification accuracy on the test set.
- *Tools:* Python, TensorFlow, Keras, Scikit-learn, OpenCV.

Accepted Conference Papers and Presentations

Jun – Dec 2025 | **R Chahana**, M.S. Anjali, Jayasri N.V, Nimisha R, Krishna Priya G. Sentiment Analysis in Finbert for Stock Market Text Data using Global News Data. *Presented at the 16th International Conference on Cloud Computing, Data Science & Engineering (CONFLUENCE), Amity University, 2026* (forthcoming in Springer proceedings). (Presentation Certificate)

Jun – Dec 2025 | M.S. Anjali, **R Chahana**, Jayasri N.V, Nimisha R, Krishna Priya G. Enhanced Stock Prediction Using a Hybrid LSTM-CNN Algorithm and Explainable AI with YFinance Data. *International Conference on Trusted Networks and Intelligent Systems (TrustNet), Manipal University Jaipur, 2026* (forthcoming in Springer proceedings). (Presentation Certificate) (presented by a co-author)

Technical Skills

Programming & Databases | Python, Java, C/C++, R, SQL, JavaScript, HTML, CSS, MySQL, MongoDB, Cassandra

AI / ML | Scikit-learn, PyTorch, TensorFlow, AutoML, LLMs, LangChain, LIME, SpaCy, NLTK, NumPy, Pandas, Polars, Matplotlib, Seaborn, BeautifulSoup, Crawl4AI

Tools & Platforms | Hadoop, Apache Pig, Flask, FastAPI, Django, Selenium, UiPath, Linux, Git, GitHub, Jupyter, Google Colab, Kaggle, GCP, Azure, Docker, Power BI, Tableau, KNIME, MS Excel, PowerPoint

Work Experience

Jun 2026 – Present

Upnex Solutions – Remote (USA)
AI Automation Intern (Part-time) | Offer Letter

- Developing an AI-powered Maintenance Follow-up Agent for US property management, automating vendor assignment, follow-ups, workflow tracking and status updates via the Buildium and LeadSimple APIs.
- Building LLM-based automation workflows to streamline vendor communication and reduce manual intervention.
- *Tools:* Python, n8n, REST APIs, Postman, Large Language Models (LLMs),Google Cloud Platform (GCP), Docker, Model Context Protocol (MCP), Render.

Apr 2025 – Jun 2025

Kambaa Incorporation – Coimbatore, India
AI Developer Intern | Internship Completion Certificate

- Developed a personal AI assistant prototype with automated MS Teams meeting notes, email draft generation and voice-based event scheduling in Outlook and Gmail Calendar.
- Implemented secure API integration (OAuth, API keys, tokens) and contextual LLM interactions using Model Context Protocol (MCP).
- *Tools:* Python, Django, Postman, Whisper, Wispr Flow, OAuth, API keys, tokens, LLMs, MCP.

Jun 2024– Jul 2024

Larsen & Toubro Limited – Coimbatore, India
Data Science Intern | Internship Completion Certificate

- Built a Power BI dashboard with detailed financial insights, adopted by 85% of the IT department.
- Automated processes using Robotic Process Automation (UiPath) and web scraping.
- *Tools:* MS Excel, Power BI, Robotic Process Automation (RPA), UiPath automation,

Honors and Competitions

Sep 2025

Smart India Internal Hackathon 2025 – Amrita Vishwa Vidyapeetham, Coimbatore, India
Shortlisted: top 30 out of 321 teams

Mar 2024

Hacktrix 2.0 Hackathon (Generative AI) – Sri Krishna College of Technology, Coimbatore, India
Finalist: top 10 teams | Certificate of Participation

Certifications

- Data Analytics with Excel – CIR, Amrita [Link]
- AI/ML for Geodata Analysis – IIRS, ISRO [Link]
- Fundamental Algorithm Design and Analysis (Elite) – NPTEL [Link]
- Developing LLM Applications with LangChain – Zero To Mastery [Link]

Languages

English (professional working proficiency), German (elementary proficiency), Tamil (professional working proficiency), Telugu (native)